MSG No. 205.34 $Pa\bar{3}1'$ [Type II, cubic]

Table 1: Wyckoff site: `4a`, site symmetry: `.-3'.1'`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,5,6,13,17,18,25,29,30,37,41,42]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[2,10,11,14,22,23,26,34,35,38,46,47]`
3	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,7,12,15,19,24,27,31,36,39,43,48]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[4,8,9,16,20,21,28,32,33,40,44,45]`

Table 2: Wyckoff site: `4b`, site symmetry: `.-3'.1'`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[1,5,6,13,17,18,25,29,30,37,41,42]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[2,10,11,14,22,23,26,34,35,38,46,47]`
3	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[3,7,12,15,19,24,27,31,36,39,43,48]`
4	$\left[0,\ \frac{1}{2},\ 0\right]$	`[4,8,9,16,20,21,28,32,33,40,44,45]`

Table 3: Wyckoff site: `8c`, site symmetry: `.3.1'`

No.	position	mapping
1	$[x,\ x,\ x]$	`[1,5,6,25,29,30]`
2	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ -x\right]$	`[2,10,11,26,34,35]`
3	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[3,7,12,27,31,36]`
4	$\left[\frac{1}{2}-x,\ -x,\ x+\frac{1}{2}\right]$	`[4,8,9,28,32,33]`
5	$[-x,\ -x,\ -x]$	`[13,17,18,37,41,42]`
6	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ x\right]$	`[14,22,23,38,46,47]`
7	$\left[x,\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$	`[15,19,24,39,43,48]`
8	$\left[x+\frac{1}{2},\ x,\ \frac{1}{2}-x\right]$	`[16,20,21,40,44,45]`

Table 4: Wyckoff site: `24d`, site symmetry: `11'`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1,25]`
2	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	`[2,26]`
3	$\left[-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[3,27]`
4	$\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$	`[4,28]`
5	$[z,\ x,\ y]$	`[5,29]`
6	$[y,\ z,\ x]$	`[6,30]`
7	$\left[-y,\ z+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[7,31]`
8	$\left[\frac{1}{2}-z,\ -x,\ y+\frac{1}{2}\right]$	`[8,32]`
9	$\left[\frac{1}{2}-y,\ -z,\ x+\frac{1}{2}\right]$	`[9,33]`
10	$\left[z+\frac{1}{2},\ \frac{1}{2}-x,\ -y\right]$	`[10,34]`
11	$\left[y+\frac{1}{2},\ \frac{1}{2}-z,\ -x\right]$	`[11,35]`
12	$\left[-z,\ x+\frac{1}{2},\ \frac{1}{2}-y\right]$	`[12,36]`
13	$[-x,\ -y,\ -z]$	`[13,37]`

continued ...

Table 4

No.	position	mapping
14	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$	[14,38]
15	$\left[x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[15,39]
16	$\left[x+\frac{1}{2},\, y,\, \frac{1}{2}-z\right]$	[16,40]
17	$[-z,\, -x,\, -y]$	[17,41]
18	$[-y,\, -z,\, -x]$	[18,42]
19	$\left[y,\, \frac{1}{2}-z,\, x+\frac{1}{2}\right]$	[19,43]
20	$\left[z+\frac{1}{2},\, x,\, \frac{1}{2}-y\right]$	[20,44]
21	$\left[y+\frac{1}{2},\, z,\, \frac{1}{2}-x\right]$	[21,45]
22	$\left[\frac{1}{2}-z,\, x+\frac{1}{2},\, y\right]$	[22,46]
23	$\left[\frac{1}{2}-y,\, z+\frac{1}{2},\, x\right]$	[23,47]
24	$\left[z,\, \frac{1}{2}-x,\, y+\frac{1}{2}\right]$	[24,48]